

SCIENCE PRO

CLASS 10 · PHYSICS · CHAPTER 1

Sound Waves — PYQs

This is a SAMPLE set of exam-style questions, written to demonstrate the download feature. Replace this file with your actual previous year question paper via VSCode — keep the same filename so the site picks it up automatically.

Section A — One-mark questions

- 1 Define frequency of a wave.
- 2 What is the SI unit of wavelength?
- 3 State the factor on which the loudness of a sound depends.
- 4 What is meant by persistence of hearing?

Section B — Two-mark questions

- 1 Distinguish between frequency and time period of a wave.
- 2 Explain why pitch depends on frequency but not on amplitude.
- 3 Why is an echo not heard in a small room?

Section C — Numerical problems

- 1 The frequency of a source of sound is 500 Hz. Find its time period.
- 2 A wave has a wavelength of 0.68 m and travels at 340 m/s. Calculate its frequency.
- 3 The echo of a sound is heard 0.8 s after it is produced. If the speed of sound in air is 350 m/s, find the distance of the reflecting surface from the source.
- 4 Calculate the minimum distance required between a source of sound and a reflecting surface for an echo to be heard distinctly, given the speed of sound in air is 340 m/s and the persistence of hearing is 0.1 s.
- 5 A ship sends a sound signal downward and receives its echo from the sea bed after 1.4 s. If the speed of sound in water is 1500 m/s, find the depth of the sea at that point.

Section D — Short answer / explain

- 1 A firecracker's echo is heard by the person who burst it 1 second later. Explain, with a calculation, how far the reflecting surface must be (speed of sound in air = 350 m/s).
- 2 Explain why the speed of sound is different in air, water and steel.